

CAB420 Assessment 1B — Question 1

1. Pre-processing

The original dataset consisted of RGB person images with a resolution of 128×64 pixels. To ensure a fair comparison between the non-deep learning and deep learning approaches, identical pre-processing was applied to both methods. Images were resized to 64×32 pixels, converted to greyscale, and normalised to the range [0,1].

This pre-processing was selected primarily to reduce computational complexity while retaining the key structural information required for person matching, including body silhouette, clothing shape, and general appearance. Converting the images to greyscale substantially reduced the dimensionality of the input data, which was particularly important for the PCA + LDA method due to the high dimensionality of flattened image vectors. Using smaller images also reduced memory usage and training time for both approaches.

However, these choices introduce trade-offs. While greyscale conversion improves efficiency, it removes colour information that is often highly discriminative in person re-identification tasks, particularly clothing colour. As a result, the selected pre-processing improves computational efficiency at the cost of potentially reduced retrieval accuracy.

After pre-processing, the training set contained 4,711 images from 250 identities. The validation set contained 50 gallery/probe identity pairs, while the final test set consisted of 301 gallery images and 301 probe images. Identity labels were extracted directly from the filenames, where the first four digits corresponded to the person identity.

2. Methods

2.1 Non-deep learning method: PCA + LDA

The non-deep learning method used PCA followed by LDA. Each pre-processed image was flattened into a vector. The training data was standardised, then PCA was fitted to reduce the image vectors to 32 dimensions. The selected PCA representation retained approximately **66.83%** of the variance.

PCA was used before LDA because the original flattened image vectors were high-dimensional relative to the number of training samples. Reducing the feature space first makes the later LDA step more stable. LDA was then trained using the identity labels from the training set. Unlike PCA, which preserves directions of high variance, LDA uses class labels and seeks a projection that improves separation between identities. The final PCA + LDA features were L2-normalised, and Euclidean distance was used to compare each probe image with all gallery images.

This approach is simple, however, it operates on flattened pixel values, so it is sensitive to changes in pose, camera viewpoint, background, lighting and alignment. This means it may work well when the probe and gallery images look visually similar, but struggle when the same person appears from a different angle or under different conditions. In addition, the greyscale pre-processing of the data, makes it more difficult for the model to find matches, due to the missing colour information.

2.2 Deep learning method: Triplet CNN with triplet loss

The deep learning method used a Triplet CNN trained with triplet loss. During training, each input consisted of three images:

- an anchor image,

- a positive image of the same identity,
- and a negative image of a different identity.

A total of 9,422 triplets were generated from the labelled training data, which was approximately twice the number of training images.

The model was trained to learn an embedding space where images of the same person were placed closer together, while images of different people were placed further apart. After training, the shared CNN backbone was used to extract feature embeddings from both the probe and gallery images. Retrieval was then performed by comparing Euclidean distances between embeddings and ranking the gallery images from most similar to least similar.

This approach is more suitable for person re-identification than a standard classifier because the identities in the test set were not seen during training. Instead of learning to classify specific people, the Triplet CNN learned general visual similarity patterns that could be applied to unseen identities.

3. Evaluation and Comparison

Evaluation was performed using Top-1, Top-5, Top-10 accuracy and CMC curves. For each probe image, every gallery image was ranked from closest to furthest based on Euclidean distance in the learned feature space. Top-1 accuracy measures whether the correct identity was ranked first. Top-5 and Top-10 measure whether the correct identity appeared within the first five or ten ranked gallery images.

Table 1. Final test-set retrieval accuracy.

Method	Top-1	Top-5	Top-10
PCA + LDA	10.63%	22.59%	28.90%
Triplet CNN	15.28%	35.22%	48.17%

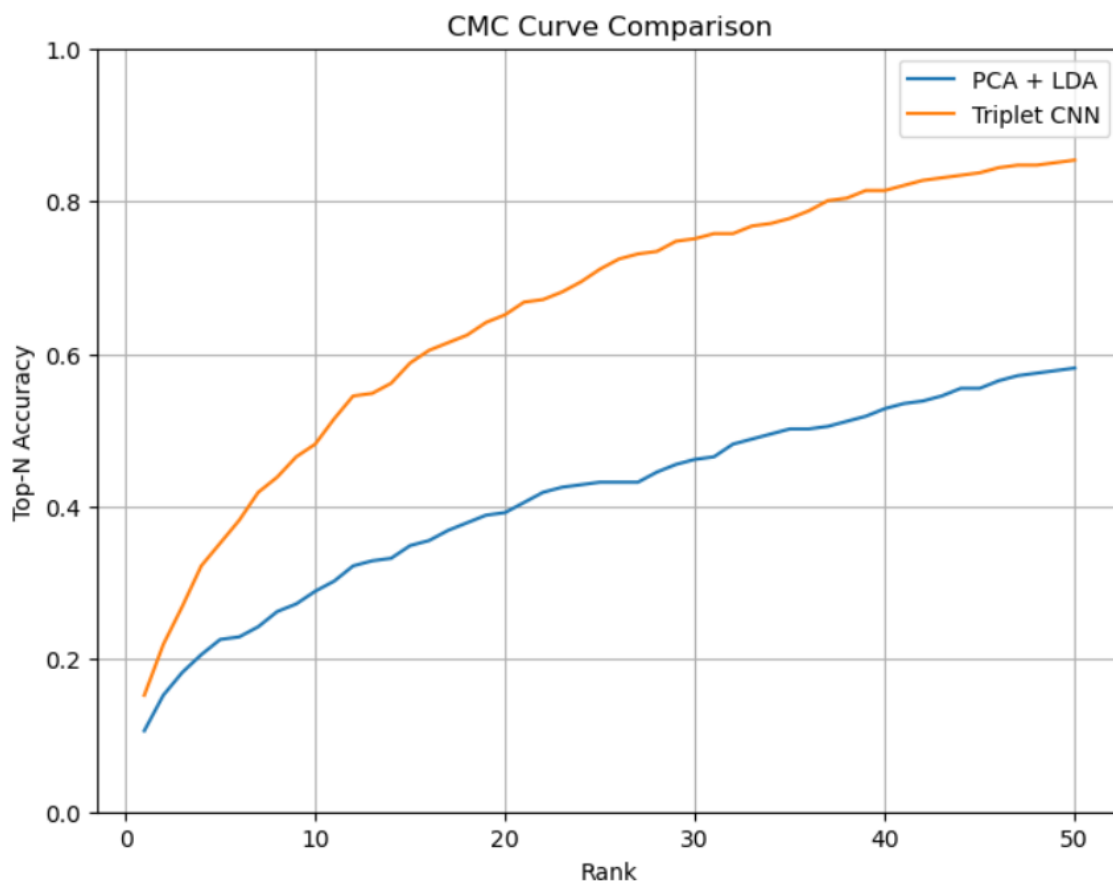


Figure 2. CMC curve comparison for PCA + LDA and Triplet CNN on the 301-identity test set.

The Triplet CNN outperformed PCA + LDA at all reported ranks. Top-1 accuracy improved from **10.63%** to **15.28%**, Top-5 improved from **22.59%** to **35.22%**, and Top-10 improved from **28.90%** to **48.17%**. The CMC curve also shows that the Triplet CNN rises much more quickly than PCA + LDA, meaning the correct identity is more often ranked near the top of the gallery list.

The rank comparison further supports this result. Across the 301 probe images, the Triplet CNN ranked the correct identity higher than PCA + LDA in **201 cases**, while PCA + LDA ranked the correct identity higher in **77 cases**, with **23 ties**. The average correct-match rank was also much lower for the Triplet CNN, improving from **58.85** for PCA + LDA to **27.37** for the Triplet CNN. The median rank improved from **35** to **11**. This shows that even when the Triplet CNN did not place the correct identity first, it usually placed it much closer to the top of the ranked list.

The PCA + LDA method remains useful because it is computationally efficient and does not require iterative neural network training. However, because it works on flattened greyscale pixel patterns, it struggles with non-linear visual changes such as pose, camera angle, background and lighting. The Triplet CNN is slower to train, but it learns local visual features through convolutional layers and is trained directly for relative similarity. This explains why it performs better overall on the retrieval task.

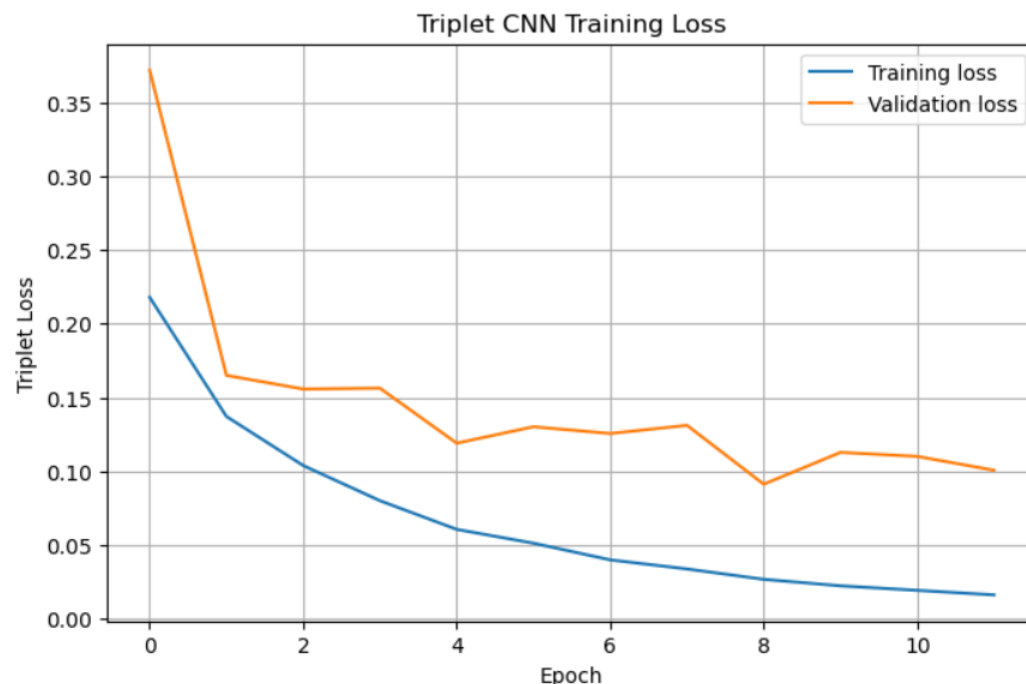


Figure 3. Triplet CNN training and validation loss over 12 epochs.

The Triplet CNN training loss decreased steadily from approximately **0.218** to **0.016**, while the validation loss decreased from approximately **0.372** to around **0.101**. The gap between training and validation loss indicates some overfitting, which is expected because the dataset is limited and the triplets were randomly generated. However, the validation loss remained much lower than at the start of training, and the final retrieval results show that the learned embedding generalised better than PCA + LDA.

4. Ethical Considerations

Person re-identification systems raise important ethical concerns because they can be used to identify or track people across cameras without active participation from the individual. The first concern is data collection. Even if images are captured in public spaces, people may not reasonably expect their appearance to be used to train biometric or re-identification systems. Dataset collection should therefore consider consent, transparency, purpose limitation, retention periods and whether individuals can opt out.

Deployment context is also important. Person re-identification may be useful in controlled authentication, missing-person search or safety-related applications, but it becomes more intrusive when used for broad surveillance, retail monitoring or law enforcement. People should be clearly notified when such systems are operating, what data is being collected, how long it will be stored, and how model outputs will be reviewed.

The results from this experiment also show why technical limitations must be considered. Even the stronger Triplet CNN only achieved **15.28% Top-1 accuracy**, meaning most probe images were not matched correctly at Rank 1. In real-world deployment, false matches and missed matches could lead to unfair treatment if the system output is treated as certain. These systems should therefore include human review, bias testing, strict access controls, clear retention limits, and a justification that less intrusive alternatives are insufficient.

CAB420 Assessment 1B — Question 2

1. Pre-processing

The original images vary in size, so all images were resized to **128×128 pixels**. This resolution was selected as a compromise between preserving enough spatial detail for segmentation and reducing the memory and training time required by both models.

The images were kept as RGB rather than converted to greyscale because breed classification can depend on colour and fur-pattern information. This was also important for the fine-tuned MobileNetV3Small model, since MobileNetV3Small was originally trained on RGB ImageNet images. Pixel values were normalised using MobileNet-style scaling to approximately $[-1, 1]$, which improves neural network training stability.

The segmentation masks were converted into a binary foreground/background format, where the pet and boundary pixels were treated as foreground and the remaining pixels were treated as background. When resizing the masks, nearest-neighbour interpolation was used rather than bilinear interpolation. This was necessary because segmentation masks are class labels, not continuous images, so interpolation should not create invalid intermediate label values.

The same pre-processing was applied to both the from-scratch model and the MobileNetV3Small model to make the comparison fair. A batch size of **16** was used due to compute restrictions. Horizontal flip augmentation was applied to the training set only because flipping a cat or dog image preserves both the breed label and the foreground/background segmentation structure. Validation and test data were not augmented.

2. Methods

2.1 From-scratch multi-task CNN

The first model was trained from scratch using an encoder-decoder architecture with one shared image input and two outputs: breed classification and semantic segmentation. The encoder used repeated convolution, batch normalisation, ReLU activation, and max pooling layers to progressively reduce the spatial size of the image while learning higher-level visual features.

The shared bottleneck representation was then used by two task-specific heads. The classification head used global average pooling, a dense layer, dropout, and a final softmax layer with 37 outputs corresponding to the pet breeds. The segmentation head used an upsampling decoder with skip connections from earlier encoder layers to reconstruct a 128×128×1 foreground/background mask. The skip connections helped preserve spatial detail required for accurate boundary prediction.

The model was trained using sparse categorical cross entropy for classification and binary cross entropy for segmentation. The two losses were combined with equal weight. Adam optimisation was used with an initial learning rate of 0.001, together with early stopping and learning-rate reduction based on validation loss.

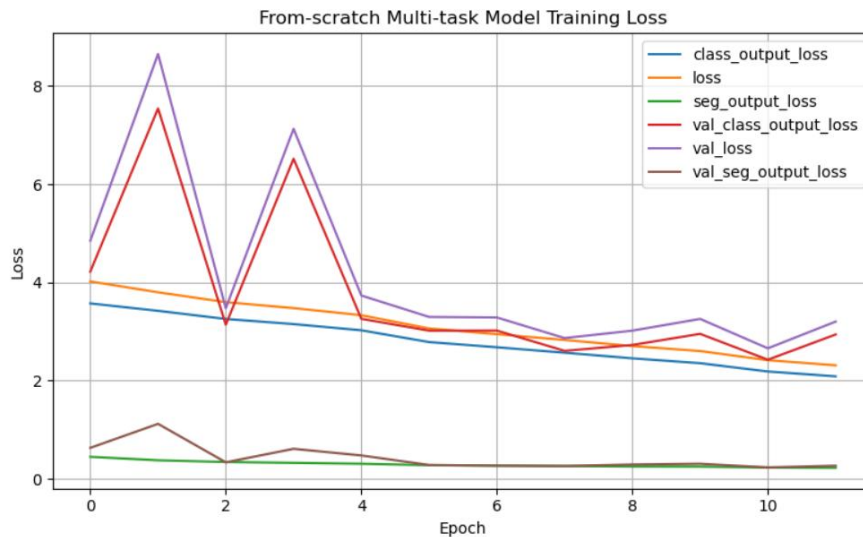


Figure 1. From-scratch multi-task model training loss.

The training loss decreased steadily, while the validation loss fluctuated early in training. This suggests that the model initially struggled to generalise because all visual features had to be learned directly from the training data.

2.2 Fine-tuned MobileNetV3Small multi-task model

The second model was a fine-tuned version MobileNetV3Small backbone pre-trained on ImageNet. The original classification head was removed, and two new task-specific heads were attached: a 37-class breed classification head and a segmentation decoder that reconstructed a binary foreground/background mask.

Training was performed in two stages. First, the MobileNetV3Small backbone was frozen and only the new classification and segmentation heads were trained. This allowed the new layers to adapt to the task while preserving the pre-trained features. Second, the final layers of the backbone were unfrozen and fine-tuned using a smaller learning rate of 0.00001. Batch normalisation layers remained frozen during fine-tuning to improve stability.

Compared with the from-scratch model, MobileNetV3Small starts with pre-trained visual features such as edges, textures, and object shapes. This is expected to improve breed classification because the model does not need to learn all low-level visual features from the training data alone.

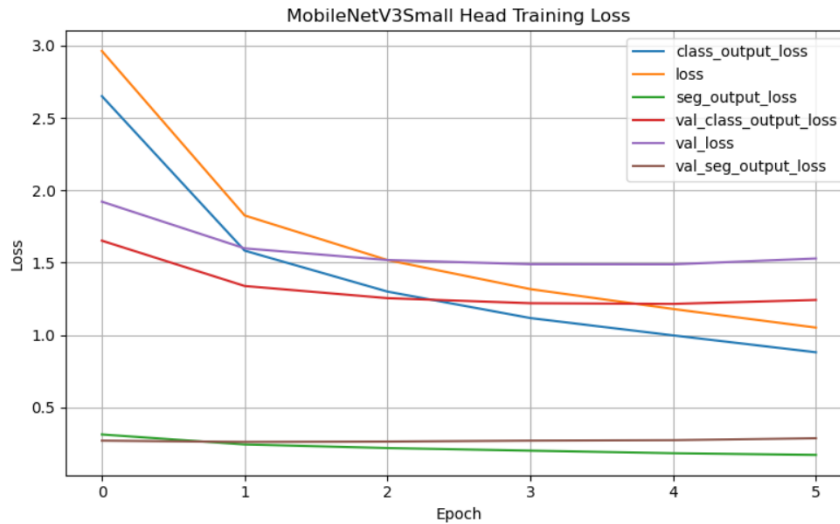


Figure 2. MobileNetV3Small head-training loss.

The head-training loss decreased quickly, indicating that the new task-specific heads were able to learn useful mappings from the frozen backbone features.

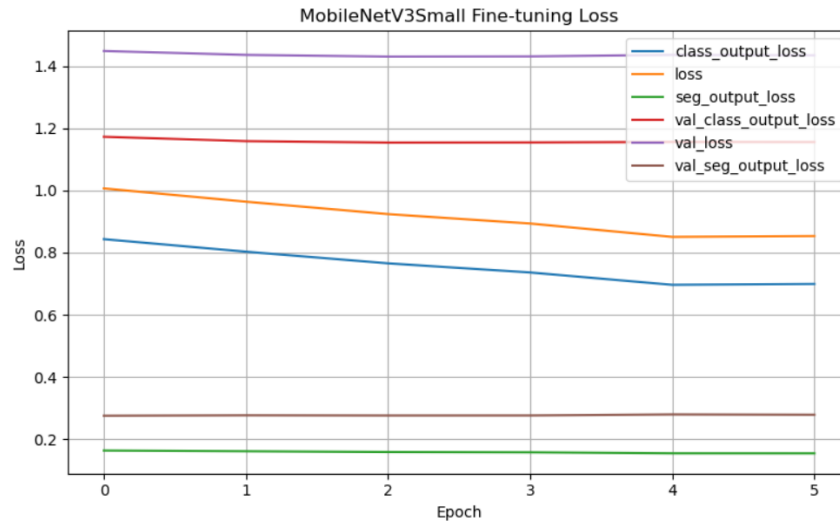


Figure 3. MobileNetV3Small fine-tuning loss.

During fine-tuning, the training loss continued to decrease gradually while the validation loss remained relatively stable. This suggests that the lower learning rate allowed the model to adapt to the task without strongly disrupting the pre-trained backbone representation.

3. Evaluation and Comparison

The final models were evaluated on both tasks using the pre-defined test split. Classification performance was measured using accuracy and macro F1. Segmentation performance was measured using pixel accuracy, foreground IoU, mean IoU and foreground Dice score.

Table 1. Final test-set performance for both multi-task models.

Model	Classification Accuracy	Macro F1	Pixel Accuracy	Foreground IoU	Mean IoU	Foreground Dice
From-scratch CNN	29.48%	0.2806	90.79%	0.8008	0.8272	0.8894
Fine-tuned MobileNetV3Small	67.25%	0.6684	89.82%	0.7841	0.8113	0.8790

The fine-tuned MobileNetV3Small model performed much better on classification, achieving **67.25% accuracy** compared with **29.48%** for the from-scratch CNN. Its macro F1 score was also much higher, increasing from **0.2806** to **0.6684**. This indicates that the pre-trained MobileNet backbone provided a much stronger representation for recognising pet breeds. The from-scratch model had to learn low-level and high-level visual features only from the Oxford-IIIT Pets dataset, which made the classification task much harder.

For semantic segmentation, the from-scratch model slightly outperformed MobileNetV3Small across all segmentation metrics. The from-scratch CNN achieved **90.79% pixel accuracy**, **0.8008 foreground IoU**, **0.8272 mean IoU** and **0.8894 foreground Dice**, compared with **89.82% pixel accuracy**, **0.7841 foreground IoU**, **0.8113 mean IoU** and **0.8790 foreground Dice** for MobileNetV3Small. This result is reasonable because the from-scratch model used skip connections in its decoder, allowing early spatial information to be reused during mask reconstruction. In contrast, the MobileNetV3Small segmentation decoder was attached to a low-resolution backbone feature map, which may have made it harder to recover precise object boundaries.

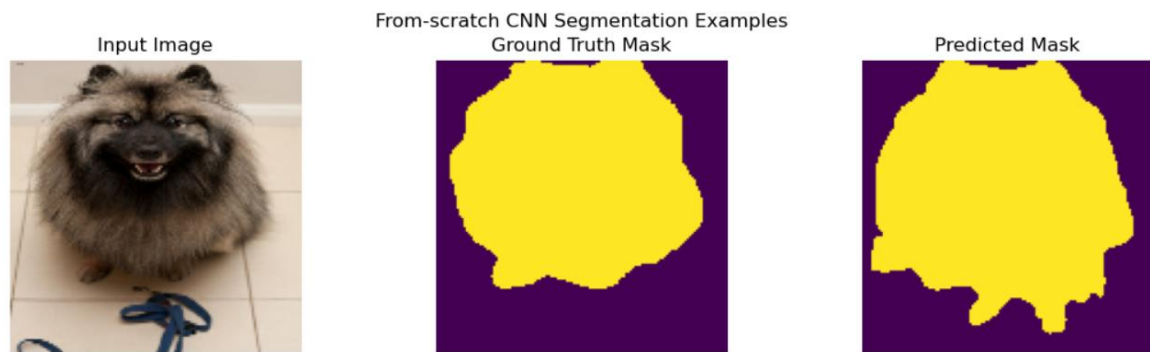


Figure 4. From-scratch CNN segmentation example.

The from-scratch CNN successfully identified the general foreground region of the dog and produced a segmentation mask that closely matched the ground truth mask. The predicted mask preserved most of the object shape, although some boundary regions around the legs and lower body were less precise.

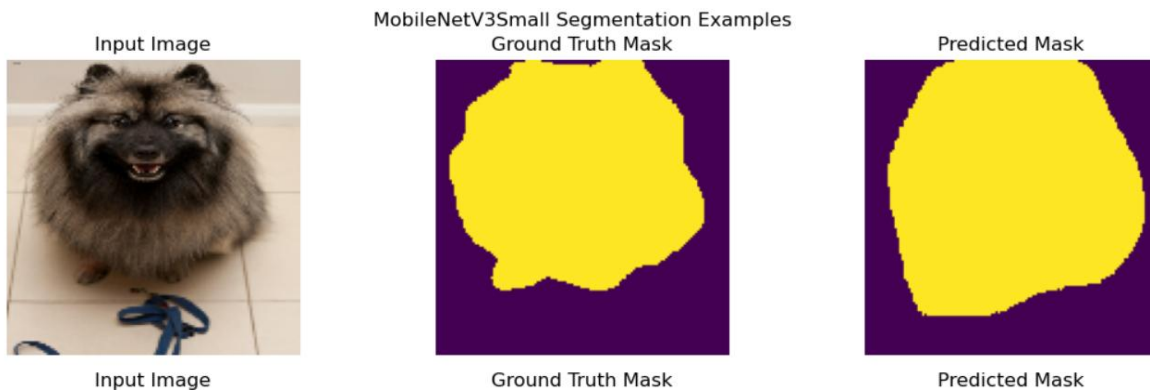


Figure 5. MobileNetV3Small segmentation example.

The MobileNetV3Small model correctly segmented the main foreground region of the dog, but the predicted mask was smoother and less detailed than the ground truth mask. Fine boundary details around the lower body and legs were less accurately reconstructed compared with the from-scratch CNN.

Overall, MobileNetV3Small was clearly stronger for classification, while the from-scratch model was slightly stronger for segmentation. This shows that the two tasks benefited from different aspects of the model design. Classification benefited most from pre-trained high-level visual features, while segmentation benefited from retaining and reusing spatial information through the decoder.

4. Improvement Attempts and Discussion

Several strategies were explored to improve performance and mitigate the main issues observed during training.

First, the input size was set to **128×128 pixels**. A larger image size may preserve more detail for segmentation and breed classification, but it would also significantly increase training time and memory usage. A smaller size, such as 64×64, would train faster but would likely remove fine fur, face and boundary details. Therefore, 128×128 was used as a practical compromise.

Second, horizontal flip augmentation was applied to the training data. This was chosen because it increases pose variation while preserving the class label and segmentation mask. More aggressive augmentation, such as strong rotation, random cropping or heavy colour jitter, was not used in the final version because these transformations could remove important pet features or make the segmentation mask less aligned with the image. In future work, carefully controlled colour augmentation could be tested because it may improve robustness without destroying the breed label.

Third, dropout and learning-rate reduction were used to reduce overfitting. The from-scratch model showed unstable validation loss during training, especially early in the process. This is expected because the model has no pre-trained knowledge and must learn all visual features from the dataset. The training curve shows that the training loss gradually decreased, but the validation loss fluctuated, suggesting that the model was sensitive to the training data and did not generalise as reliably as MobileNetV3Small.

Fourth, MobileNetV3Small was trained in two stages. Freezing the backbone first allowed the newly added heads to learn useful task-specific mappings before any pre-trained features were modified. Fine-tuning the final backbone layers with a very small learning rate then allowed the model to adapt slightly to cats, dogs and segmentation without destroying the useful ImageNet representation. The MobileNet head-training and fine-tuning loss curves show that the model trained more smoothly than the from-scratch CNN, especially for classification.

A possible improvement for the MobileNetV3Small segmentation task would be to use a stronger encoder-decoder segmentation head with skip connections from intermediate MobileNet layers. This would allow the decoder to access higher-resolution spatial features rather than relying mainly on the final low-resolution feature map. This may improve boundary detail around legs, tails, fur and small animal regions. Other possible improvements include using class weighting or focal loss for difficult breeds, testing a higher input resolution such as 160×160 or 192×192, and training for longer with stronger regularisation. These were not implemented because they would increase computational cost and model complexity.

The results show that the fine-tuned model was much better for classification, while the from-scratch model remained competitive for segmentation. This supports the idea that pre-trained backbones are valuable when labelled data and compute are limited, but segmentation performance also depends heavily on decoder design and the ability to recover spatial detail.